

7.1 Waves: Transverse & Longitudinal

Question Paper

Course	CIEA Level Physics
Section	7. Waves
Topic	7.1 Waves: Transverse & Longitudinal
Difficulty	Hard

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The equation below shows the speed of v waves in deep water

$$v^2 = \frac{g\lambda}{2\pi}$$

Where g is the acceleration of free fall and λ is the wavelength of the waves. The wavelength λ and the frequency f of the wave were measured.

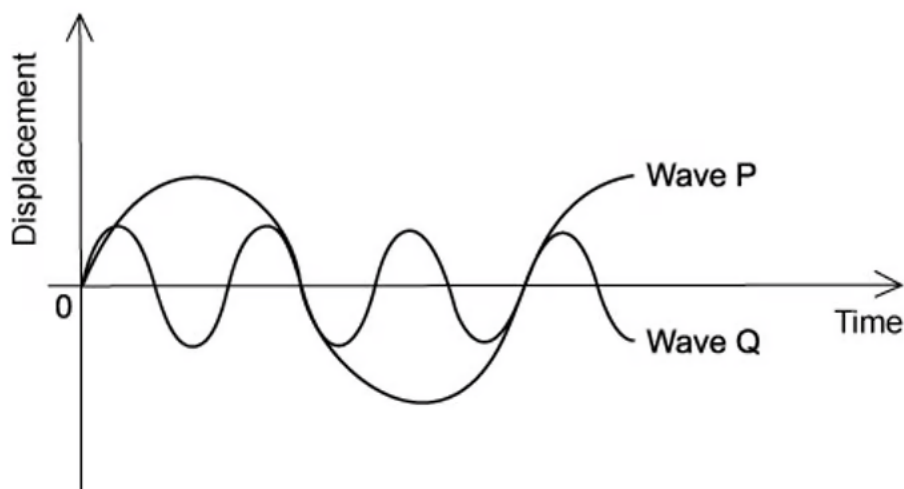
If the graph was to give a straight line through the origin which of the following graphs should be plotted?

- A. f against λ^2
- B. f^2 against λ
- C. f against $\frac{1}{\lambda}$
- D. f^2 against $\frac{1}{\lambda}$

[1 mark]

Question 2

The graph shows two waves P and Q.



Wave P has an amplitude of 8 cm and a frequency 100 Hz.

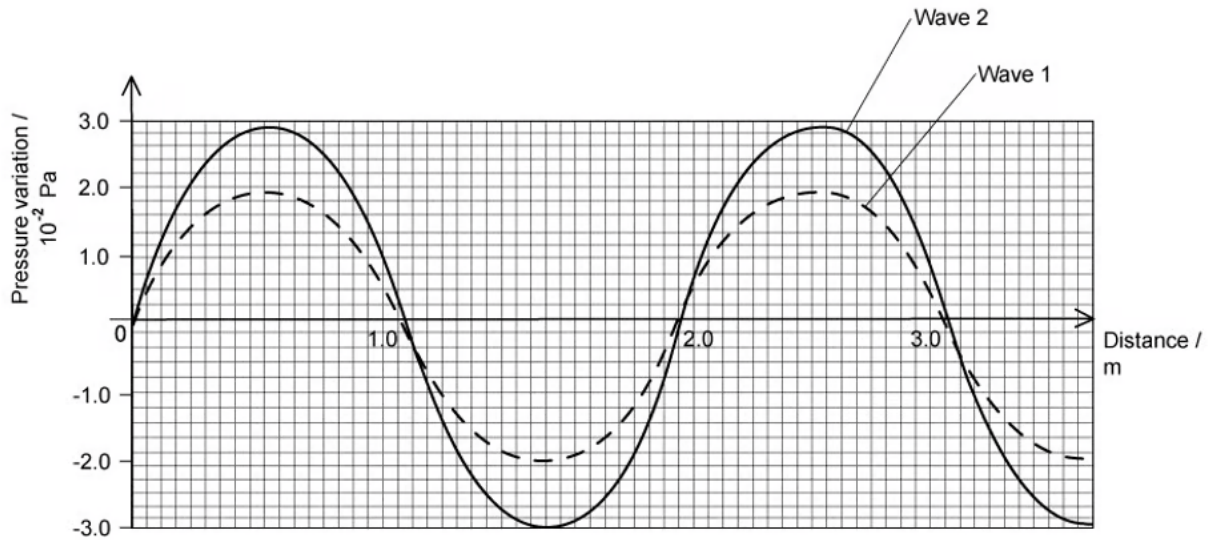
What is the amplitude and the frequency of wave Q?

	Amplitude / cm	Frequency / Hz
A	4	300
B	4	33
C	2	300
D	2	33

[1 mark]

Question 3

The graph below shows two sound waves, the sound wave is a series of moving pressure variations.



Wave 1 has an intensity of $1.6 \times 10^{-6} \text{ W m}^{-2}$, what is the intensity of wave 2?

- A. $3.6 \times 10^{-6} \text{ W m}^{-2}$
- B. $2.4 \times 10^{-6} \text{ W m}^{-2}$
- C. $3.0 \times 10^{-6} \text{ W m}^{-2}$
- D. $4.5 \times 10^{-6} \text{ W m}^{-2}$

[1 mark]

Question 4

A light wave is incident normally on a surface of area S with an amplitude A . The power per unit area reaching the surface is P .

If the light is focused on a smaller area of $\frac{1}{3} S$, and the amplitude of wave is doubled to $2A$ what is the power per unit area?

- A. $6P$
- B. $12P$
- C. $18P$
- D. $36P$

[1 mark]

Question 5

All electromagnetic waves travel at speed c in a vacuum with a frequency f and wavelength λ .

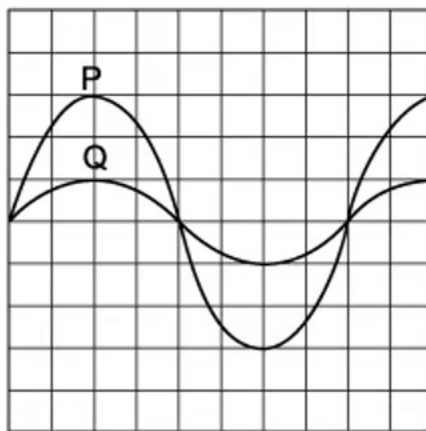
Which row in the table describes the wavelength and speed of electromagnetic waves with a frequency $\frac{f}{2}$?

	Speed in a vacuum	Wavelength
A	$2c$	2λ
B	c	2λ
C	c	$\frac{\lambda}{2}$
D	$\frac{c}{2}$	$\frac{\lambda}{2}$

[1 mark]

Question 6

Sound waves can be displayed on a cathode ray oscilloscope (C.R.O.). The graph below represents two waves seen on the screen. They are labelled P and Q as shown in the diagram



What is the ratio $\frac{\text{intensity of sound wave P}}{\text{intensity of sound wave Q}}$?

A. $\frac{\sqrt{3}}{1}$

B. $\frac{9}{1}$

C. $\frac{1}{1}$

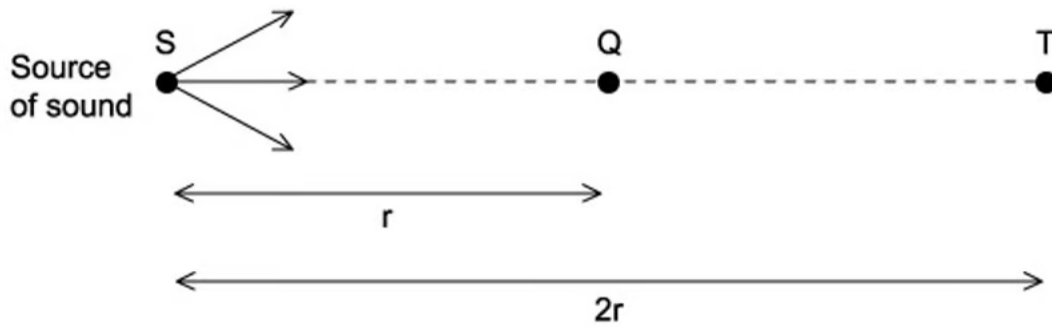
D. $\frac{3}{1}$

[1 mark]

Question 7

The intensity I of a sound at point Q is inversely proportional to the square of the distance x of Q from the source of the sound. This can be shown by the equation

$$I \propto \frac{1}{x^2}$$



Molecules of air at Q are a distance r from S and oscillate with amplitude $8.0 \mu\text{m}$.

Point T is a distance $2r$ from S.

What is the amplitude of oscillation of air molecules at T?

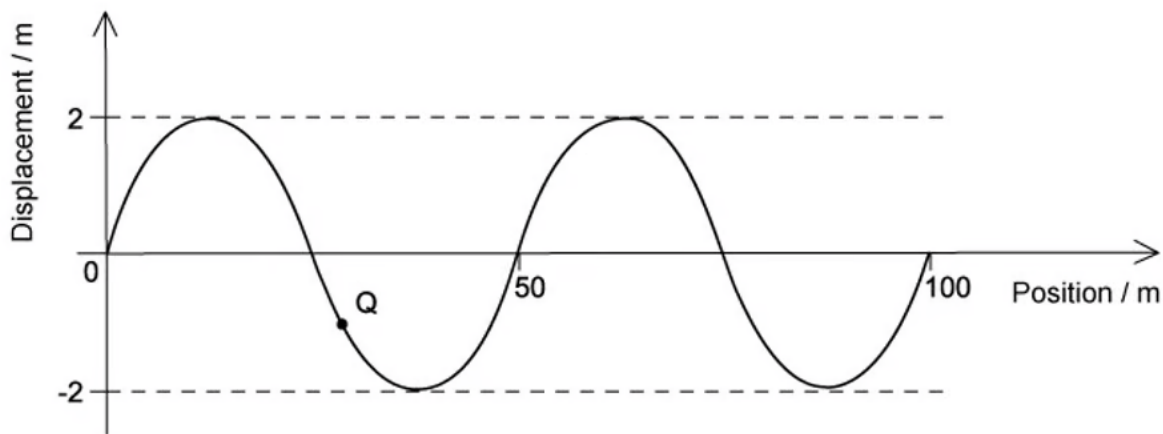
- A. $4.0 \mu\text{m}$
- B. $2.0 \mu\text{m}$
- C. $2.8 \mu\text{m}$
- D. $1.4 \mu\text{m}$

[1 mark]

Question 8

The graph shows a sinusoidal wave in water, travelling at a speed of 8.0 m s^{-1}

The maximum speed of the particles in the wave is where a is the amplitude and f is the frequency.



An object Q is floating on the water it has a mass of $2.5 \times 10^{-3} \text{ kg}$

What is the maximum kinetic energy of Q due to the wave? Assume the motion is vertical.

- A. 4.1 mJ
- B. 5.1 mJ
- C. 39 mJ
- D. 64 mJ

[1 mark]

Question 9

A wave has a speed of 340 m s^{-1} and a frequency of 500 Hz

If two points on a wave are 0.17 m apart, what is the phase difference?

- A. $\pi \text{ rad}$
- B. $\frac{3\pi}{4} \text{ rad}$
- C. $\frac{\pi}{4} \text{ rad}$
- D. $\frac{\pi}{2} \text{ rad}$

[1 mark]

Question 10

The sun emits light of a particular wavelength λ_s . At any instant, there is a range of wavelengths from less than λ_s to greater than λ_s that can be observed on Earth. This is due to the Doppler effect.

Which of the following statements would explain this Doppler effect?

- A. the Sun is rotating
- B. the Sun is moving toward the Earth
- C. the Sun is moving away from the Earth
- D. the Sun is moving at right-angles to a line joining the Sun and the Earth

[1 mark]

Model Answers: Hard

1

The correct answer is **D** because:

- The equation given in the question is $v^2 = \frac{g\lambda}{2\pi}$
- The equation for the speed of the wave is $v = f\lambda$
- By substituting $v = f\lambda$ into the first equation gives
 - $(f\lambda)^2 = \frac{g\lambda}{2\pi}$
 - $f^2 = \left(\frac{g\lambda}{2\pi}\right)\left(\frac{1}{\lambda^2}\right) = \left(\frac{g}{2\pi}\right)\left(\frac{1}{\lambda}\right)$
- The equation of a straight line is $y = mx + c$
- A line going through the origin has a y-intercept of $c = 0$ so $y = mx$
 - Comparing with $f^2 = \left(\frac{g}{2\pi}\right)\left(\frac{1}{\lambda}\right)$ with $y = mx$
- We can see y is equal to f^2 and x is equal to $\frac{1}{\lambda}$, so, a plot of these two quantities on a graph would give a straight line through the origin

2

The correct answer is **A** because:

- The amplitude is the maximum displacement of the wave
- The amplitude of P is 8 cm and from the diagram we can see wave Q has an amplitude of half that so 4 cm
- Frequency is the number of waves per second of the wave
- Wave P has a frequency of 100 Hz, for every 1 wave that passes of P there are 3 waves of Q
- The period is the time it takes for a wave to complete one wavelength
 - So, the period of P = $\frac{1}{100} = 0.01 \text{ s}$

- The period of Q is $= \frac{0.01}{3}$
- The frequency of Q $= \frac{1}{\text{period}} = \frac{3}{0.01} = 300 \text{ Hz}$
 - Therefore, wave Q has an amplitude of 4 cm and a frequency of 300 Hz

3

The correct answer is **A** because:

- Intensity I is proportional to amplitude squared A^2 , i.e.
 - $A^2 \propto I$
 - $\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$
- From the graph we can see the amplitude and intensity of wave 1 is equal to
 - $A_1 = 2$
 - $I_1 = 2^2 = 4$
- This corresponds to an intensity of $1.6 \times 10^{-6} \text{ W m}^{-2}$
- From the graph we can see the amplitude and intensity of wave 2 is equal to
 - $A_2 = 3$
 - $I_2 = 3^2 = 9$
- The ratios of I_1 and I_2 can be used to find the corresponding intensity of wave 2
 - $\frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$
 - $\frac{1.6 \times 10^{-6}}{I_2} = \frac{4}{9}$
 - $I_2 = \frac{9}{4} \times (1.6 \times 10^{-6}) = 3.6 \times 10^{-6}$
- Therefore, the intensity of wave 2 is $3.6 \times 10^{-6} \text{ W m}^{-2}$

4

The correct answer is **B** because:

- The power per unit area = $\frac{\text{power}}{\text{area}}$ intensity
- When the surface area is S and the amplitude is A , the power per unit area = P
- The surface area has decreased to $\frac{1}{3} S$, since intensity is inversely proportional to the area, i.e.
 - $I \propto \frac{1}{\frac{1}{3}S} = \frac{3}{S}$
- So, when the surface area is decreased by a factor of 3 the intensity is increased by a factor of 3
- The second factor in the question is the amplitude of the wave, which is doubled to $2A$
- Intensity is directly proportional to the square of the amplitude, i.e.
 - $I \propto A^2$
- Doubling the amplitude increases the intensity to increase by a factor of 2^2
 - $I \propto (2A)^2 = 4A^2$
- The new power will therefore be $4 \times 3 P = 12 P$

5

The correct answer is **B** because:

- The question states that the speed of the electromagnetic (EM) waves in a vacuum is c
 - Speed is given by $c = f\lambda$
 - Therefore, wavelength is equal to $\lambda = \frac{c}{f}$
- The speed of EM waves is constant in a vacuum so the speed will stay as c
- The frequency has decreased to $\frac{f}{2}$ so the new wavelength is $= \frac{c}{\frac{f}{2}} = \frac{2c}{f}$
- Comparing $\lambda = \frac{c}{f}$ with $\frac{2c}{f}$, we can see the new wavelength is $= 2\lambda$

6

The correct answer is **B** because:

- Intensity I is proportional to amplitude squared A^2
 - $I \propto A^2$
- The amplitude of P is 3 units and the amplitude of Q is 1 unit
 - The intensity of P = $3^2 = 9$
 - The intensity of Q = $1^2 = 1$
- Therefore the ratio is $\frac{\text{intensity of sound wave P}}{\text{intensity of sound wave Q}} = \frac{9}{1}$

7

The correct answer is **A** because:

- We are told intensity is inversely proportional to the square of the distance x from the source so
 - $I \propto \frac{1}{x^2}$
- When the distance is doubled this causes the intensity to decrease by a quarter
 - $I \propto \frac{1}{(2x)^2} = \frac{1}{4x^2}$
- Intensity I is also proportional to amplitude squared A^2
 - $I \propto A^2$
- From the question, we can see the amplitude is $8 \mu\text{m}$, so intensity is proportional to 8^2
 - $I \propto (8A)^2$
- So, at a distance $2r$, the intensity will be $\frac{1}{4}$ and A^2 is equal to $\frac{8^2}{4}$
- Therefore, the amplitude at T is equal to $A = \sqrt{8.0^2 \div 4} = 4.0 \mu\text{m}$

8

The correct answer is **B** because:

- The equation for the speed of a wave is $v = f\lambda$
- So, frequency is $f = \frac{v}{\lambda}$

- From the question, we can see the amplitude is $8\text{ }\mu\text{m}$, so intensity is proportional to 8^2
 - $I \propto (8A)^2$
- So, at a distance $2r$, the intensity will be $\frac{1}{4}$ and A^2 is equal to $\frac{8^2}{4}$
- Therefore, the amplitude at T is equal to $A = \sqrt{8.0^2 \div 4} = 4.0\text{ }\mu\text{m}$

8

The correct answer is **B** because:

- The equation for the speed of a wave is $v = f\lambda$
- So, frequency is $f = \frac{v}{\lambda}$
- From the graph, the wavelength is 50 m
- The question states that the speed is 8.0 m s^{-1}
 - $f = \frac{8}{50} = 0.16\text{ Hz}$
- From the graph, the amplitude is 2 m
- The equation for the maximum speed of the particles is
 - $2\pi af = 2\pi(2)(0.16) = 0.64\pi\text{ m s}^{-1}$
- Therefore, the maximum kinetic energy is equal to
 - $\frac{1}{2}mv^2 = \frac{1}{2}(2.5 \times 10^{-3})(0.64\pi)^2 = 0.0051 = 5.1\text{ mJ}$

9

The correct answer is **D** because:

- The equation for the speed of a wave $v = f\lambda$
- So, wavelength is $\lambda = \frac{v}{f} = \frac{340}{500} = 0.68\text{ m}$
- One wavelength λ is equivalent to a phase difference of 360 degrees (which is equal to 2π rad)
- The waves are in phase as they are separated by a distance equal to λ
 - The phase difference is therefore 2π rad
- Therefore, a separation of 0.68 m (one wavelength) corresponds to a phase difference of 2π rad

- So, a separation of 0.17m corresponds to $= \left(\frac{0.17}{0.68} \right) \times 2\pi = \frac{\pi}{2} \text{ rad}$

10

The correct answer is **A** because:

- The Doppler effect is when the observed frequency/wavelength of light is different to the emitted frequency/wavelength of the light due to the motion of either the source, the observer or both
- A shorter wavelength is seen when the source approaches the observer in this case the Sun moving towards the Earth
- A longer wavelength is seen when the source moves away from the observer, in this case the Sun moving away from the Earth
- One side of the Earth will observe the Sun moving away and one side will observe the Sun moving towards the Earth
- This means that from the Earth both long and short wavelengths can be seen at any instant

B is incorrect because a shorter wavelength is observed when the Sun (source) is moving towards the observer (the Earth), the question states that there is a range of wavelengths observed. So, both longer and shorter wavelengths will be seen, not only shorter as in this case

C is incorrect because a longer wavelength is observed when the Sun (source) is moving towards the observer (the Earth) the question states that there is a range of wavelengths observed. So, both longer and shorter wavelengths will be seen, not only longer as in this case

D is incorrect because if the Sun was moving at right angles then both longer and shorter wavelengths would be observed but at only specific instances in time. The shorter wavelengths would be seen as the Sun is moving toward the Earth, once it reaches the line joining the Sun and Earth it will continue to be longer